

# FEDERICO BRUZZONE ✧ CURRICULUM VITAE

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Presently a PhD Candidate in Computer Science. I have a strong passion for **Programming Languages and Compilers**, and I'm currently diving into **AI (ML/DL) compilers**. For more information, just visit my personal [webpage](#).

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## ✧ SCIENTIFIC PUBLICATIONS ✧

*Workshop paper (poster) at PNAS'26, CEUR-WS* — Lorenzo Capra and Federico Bruzzone. Symmetry preservation in modular rewritable muliformalism models. In *Joint Workshop Proceedings of PNSE'26, PNAS'26, ATAED'26 and PHOCON'26*, volume 4236 of *CEUR Workshop Proceedings*. CEUR-WS.org, 2026 — [bib](#) [pdf](#) [CEUR-WS](#)

*Journal Ranked Q1 on Scimago* — Federico Bruzzone, Walter Cazzola, Matteo Brancaleoni, and Dario Pellegrino. Sink or swim: Tackling real-time asr at scale. *IEEE Transactions on Audio, Speech and Language Processing*, pages 1–11, 2026 — [bib](#) [pdf](#) [IEEE](#) [arXiv](#)

*Journal Ranked Q1 on Scimago* — Federico Bruzzone, Walter Cazzola, and Luca Favalli. Code less to code more: Streamlining language server protocol and type system development for language families. *Journal of Systems and Software*, page 112554, September 2025 — [bib](#) [pdf](#) [SpringerLink](#) [arXiv](#)

## PREPRINTS

Federico Bruzzone and Walter Cazzola. A multi-dimensional, per-pass empirical study of the llvm optimization pipeline, 2026 — [arXiv](#)

Federico Bruzzone, Walter Cazzola, and Luca Favalli. Generalized software product line extraction, 2026 — [arXiv](#)

Federico Bruzzone and Walter Cazzola. Meta-monomorphizing specializations, 2026 — [arXiv](#)

Federico Bruzzone, Walter Cazzola, and Luca Favalli. From separate compilation to sound language composition, 2026 — [arXiv](#)

Federico Bruzzone, Walter Cazzola, and Luca Favini. Prioritizing configuration relevance via compiler-based refined feature ranking, 2026 — [arXiv](#)

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## ✧ EDUCATION ✧

**2024–Present** PhD Candidate in Computer Science at the ADAPT Lab, University of Milan

**2022–2024** MSc in Computer Science at University of Milan (*110/110 cum laude*, W. Cazzola, 15/07/2024)  
*Thesis: Toward a Modular Approach for Type Systems and LSP Generation*

**2019–2022** BSc in Musical Computer Science at University of Milan (13/10/2022)

**2011–2019** Piano and Music Composition at I.S.S.M. Novara Conservatory

**2014–2019** Diploma in Computer Science and Telecommunications at E. Alessandrini

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## ✧ OPEN SOURCE CONTRIBUTIONS ✧

**2026** LLVM Lighthouse: [Fix failed tests on Darwin](#)

**2026** MLIR/LLVM: [\[mlir\]\[SCF\] Skip dynamic front-peeling for non-index loops](#)

**2026** MLIR/LLVM: [\[mlir\]\[ArmNeon\] Enable native I8MM integration testing on Darwin](#)

**2026** MLIR/LLVM: [\[mlir\]\[vector\] Verify non-unit strides on masked/expand/compress ops](#)

**2026** MLIR/LLVM: [\[mlir\]\[ArmSME\] Enable native ArmSME integration testing on Darwin](#)

**2026** MLIR/LLVM: [\[mlir\]\[ArmSME\] Reject non-unit-stride tile\\_load/tile\\_store memrefs and pass layout{IdentityLayoutMap} to matmul tests](#)

**2026** MLIR/LLVM: [\[mlir\]\[ArmNeon\] Add linalg.matmul e2e tests](#)

**2026** MLIR/LLVM: [\[TailRecElim\] Fix miscompile of returns using a non-eliminated recursive call](#)

**2026** MLIR/LLVM: [\[mlir\]\[vector\] Verify non-unit strides on masked/expand/compress ops \(reverted by #215224\)](#)

**2026** MLIR/LLVM: [\[mlir\]\[ArmNeon\] Reject masked contractions in I8MM/BFMMMA lowering patterns](#)

- 2026 MLIR/LLVM: [\[TailRecElim\]](#) Introduce support for shift accumulator optimization
- 2026 IREE: [\[GlobalOpt\]\[Flow\]\[Quant\]](#) Fix segfault on unlowered quant.uniform types
- 2026 IREE: [\[Codegen\]\[LLVMCPU\]](#) Hoist in-loop stack alloca
- 2026 IREE: [\[LLVMCPU\]](#) Fix scalable vectorization fallback on SME-only (no +sve) targets
- 2026 MLIR/LLVM: [\[mlir\]\[sparse\]](#) Avoid vectorizing non-contiguous COO coordinate loads
- 2026 MLIR/LLVM: [\[mlir\]\[ArmSVE\]](#) Fix comment inconsistency for pack\_lhs in [ArmSVE/pack-unpack-mmt4d.mlir](#) (NFC)
- 2026 MLIR/LLVM: [\[mlir\]\[vector\]](#) add consistent stride verification to masked load/store and gather/scatter ops
- 2026 MLIR/LLVM: [\[mlir\]\[ArmSME\]](#) fix f64 scalable matmul crashes in [VectorLegalizationPass](#)
- 2026 IREE: [\[LLVMCPU\]](#) Fix ArmSME requiring classic SVE, breaking SME-only targets (e.g., Apple Silicon)
- 2026 IREE: [\[LLVMCPU\]](#) Add SME lowering-strategy tests for f64 and unsupported i8 matmuls
- 2026 MLIR/LLVM: [\[mlir\]\[VectorToLLVM\]](#) add opt-in enable-gep-inbounds-nuw pass flag for [vector.load/store](#)
- 2026 MLIR/LLVM: [\[mlir\]\[MemRefToLLVM\]](#) fix incorrect nuw on GEP/mul when lowering [memref.load/store](#) with negative strides
- 2026 MLIR/LLVM: [\[NFC\]\[mlir\]\[linalg\]](#) add [toContractionDimensions](#) for healthy code reuse
- 2026 MLIR/LLVM: [\[mlir\]\[vector\]](#) extend [createReadOrMaskedRead/createWriteOrMaskedWrite](#) with permutation map support
- 2026 MLIR/LLVM: [\[mlir\]\[affine\]](#) emit [in\\_bounds](#) on [transfer\\_read/write](#) when statically provable in [affine-super-vectorize](#)
- From 2026 [The Eter Programming Language](#), a new programming language leveraging the MLIR/IREE ecosystem with tensors support.
- From 2026 Maintainer of the [LLVM Passview](#), a framework to run an incremental, reproducible study of LLVM optimization passes.
- From 2026 Maintainer of the [LLVM Pass Template](#), a C++ template project to quickly create new LLVM passes.
- From 2025 Maintainer of the [Papers on Compiler Optimizations: Analysis and Transformations](#), a curated list of scientific publications.
- From 2025 Maintainer of [scribe](#), a minimalist, opinionated LaTeX document class for technical writing and presentations.
- From 2025 Maintainer of the [Tide Compiler](#), a backend-agnostic IR and compiler framework written in **Rust**.
- 2025 [Generalizing Closeness centrality to weighted networks using Newman method](#) (issue #1384)
- 2025 Rust: [Use the type-level constant value ty::Value](#) where needed
- 2025 Rust: [Add TooGeneric](#) variant to [LayoutError](#) and emit [Unknown](#)
- 2024 Rust: [Report the note when specified in diagnostic::on\\_unimplemented](#)
- From 2024 Maintainer of the cross-platform [tgt](#) project and [tdlib-rs](#) TDLib bindings written in **Rust**.
- From 2024 [CHIP-8 STM32-Port](#): CHIP-8 and S-CHIP emulator ported to STM32 Cortex M4 microcontroller.
- From 2023 [CHIP-8 Core](#): Core CHIP-8 and S-CHIP emulator library, used as the foundation for platform ports.

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## ◆ RESEARCH ACTIVITIES ◆

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- May 2026 **Committee Member** at the Int. Conf. on Software Language Engineering ([SLE 2026](#)), *ACM*, **B on CORE**
- Dec 2025 **Reviewer** for the Eur. Conf. on Object Oriented Programming ([ECOOP 2026](#)), *ACM*, **A on CORE**
- Nov 2025 **Reviewer** for the Journal of Computer Languages ([COLA](#)), *Elsevier*, **C on CORE**
- Jun 2025 **Reviewer** for the Journal of Software and Systems Modeling ([SoSyM](#)), *Springer*, **Q1 on Scimago**
- 2–6 Jun 2025 **Participant** at [<Programming>2025](#) conference
- Apr–Jun 2025 **Committee Member** at the Int. Conf. on Software Language Engineering ([SLE 2025](#)), *ACM*, **B on CORE**
- Mar 2025 **Reviewer** for the Journal of Systems and Software ([JSS](#)), *Elsevier*, **Q1 on Scimago**
- Sep 2024 **Student Volunteer** at the Int. Conf. on Functional Programming ([ICFP 2024](#))

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## ◇ ————— DISSEMINATION ACTIVITIES ————— ◇

- Jun 2026** [A Multi-Dimensional, Per-Pass Empirical Study of the LLVM Optimization Pipeline](#) — Ph.D. Student’s Activity Report, University of Milan.
- Mar 2026** [MLIR: Scaling Compiler Infrastructure for Domain Specific Computation](#) — Presentation of the paper by Lattner et al., 2021.
- Feb 2026** [Matheuristic Variants of DSATUR for the Vertex Coloring Problem](#) — Presentation of the paper by Dupin, 2024.
- Dec 2025** [Your Optimizing Compiler is Not Optimizing Enough. To Hell With Multiple Recursions!](#) — First MuseMI meeting of the 2025/2026 season, Milan.
- Jul 2024** [P4 Compiler in SDN](#) — Presentation on the P4 compiler in Software Defined Networking.
- Jul 2024** [PhD Project](#) — *Universal Language Server Protocol and Debugger Adapter Protocol for Modular Language Workbenches*, University of Milan.
- Jul 2024** [Master’s Thesis](#) — *Toward a Modular Approach for Type Systems and LSP Generation*, University of Milan.

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## ◇ ————— TEACHING ACTIVITIES ————— ◇

### THESIS SUPERVISION

- 14/07/2026** M. Visconti, *Towards Neverlang3: A Generative Rust Approach for Resilient LL Parser Generators*, MSc, **102**
- 23/02/2026** D. Cerato, *Employing Metaprogramming: A Macro-Driven Strategy to Overcome Compiler-Imposed Limitations on Specialization*, BSc, **103**
- 10/04/2025** D. Pellegrino, *Scalable Multi-client Real-time Whisper*, BSc, **96**
- 09/04/2025** A. Longoni, *GUIDE: Graphical User Interface Development Environment*, MSc, **110L**
- 09/04/2025** L. Albani, *New Generalized Protocol For Software Product Line Extraction And Configuration*, MSc, **110L**
- 09/04/2025** G. Esposito, *Fr3D: A Framework for DAP-compatible DSL-oriented Debugging*, MSc, **110L**
- 24/02/2025** L. Favini, *RustyEx: Instrumenting rustc to Extract Feature Dependency Graphs*, BSc, **102**

### GRADUATE COURSES

- 2025–2026** Mathematical Logic (Art. 45), **BSc** in CS, University of Milan, S. Aguzzoli
- 2024–2025** Programming 1 (Art. 45), **BSc** in CS, University of Milan, L. Capra (coordinator W. Cazzola)
- 2024–2025** Mathematical Logic (Art. 45), **BSc** in CS, University of Milan, S. Aguzzoli
- 2023–2024** Mathematical Logic, **BSc** in CS, University of Milan, S. Aguzzoli
- 2023–2024** Computer Science, **BSc** in Comm. and Society, University of Milan, A. Momigliano
- 2023–2024** Programming 1, **BSc** in CS, University of Milan, A. Trentini (coordinator P. Boldi)
- 2023–2024** Programming in Python, **MSc** in Chemistry, University of Milan, M. Monga

### ADDITIONAL ACTIVITIES

- 2023–Present** **Private Tutoring** in Computer Science, Mathematics and Physics
- Nov 2024** **Collaborator** at the Bebras Challenge, ALaDDIn Lab
- Jan–Jun 2024** **Organizer** of the BSc Computer Science Laboratories, ALaDDIn Lab
- Jan–Jun 2024** **Collaborator** of the Workshops for schools, ALaDDIn Lab
- Nov 2023** **Collaborator** at the Bebras Challenge, ALaDDIn Lab

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## ◇ ————— TECHNICAL SKILLS ————— ◇

- Languages** Rust, C/C++, Python, Go, OCaml, Java, Scala, Kotlin, Erlang, Lua, Dart, PHP, HTML/CSS, SQL, Bash, TeX
- Systems/Tooling** Git, CI/CD, Docker, GDB, Valgrind, Build Systems (e.g., CMake, Cargo, Pip), Cross-language Linking, Static/Dynamic Libraries, and FFI (e.g., C bindgen)
- Area of Expertise** Compiler Construction and Optimizations, Programming Languages Design, IR Design and Implementation, Type Systems, Language Support Tools (e.g., LSP), Static Analysis, Rustc Internals, MLIR, LLVM, Parsing Techniques and Parser Generators (e.g., ANTLR)

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◇ ————— **MUSICAL ACTIVITIES** ————— ◇

- 2021–Present** **Sound Engineer and Music Producer**, in collaboration with [Believe](#) and as a [SIAE](#) member.
- 2006–Present** **Pianist and Music Composer**, composing original pieces and arrangements for piano and various ensembles
- 2019–Present** **Piano and Music Teacher**, teaching piano and music theory and composition to students of all ages and levels

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◇ ————— **GRANTS AND FELLOWSHIPS** ————— ◇

- 2024** [56th Top Github Public Contributor](#) in Italy out of 958
- 2023–2024** Scholarship for the MSc in Computer Science, awarded by the University of Milan
- 2020–2024** Scholarship for the BSc in Musical Computer Science, awarded by the University of Milan

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◇ ————— **TONGUES** ————— ◇

- Italian** Mother tongue
- English** Level CEFR B2 (SLAM at University of Milan)
- Spanish** Base (A1–A2)